Direct air capture at climate-relevant scale: evidence and conditions for regional pilot funding

International evidence review for a regional climate committee · 9 November 2026

Executive summary

Decision in brief. Direct air capture with durable carbon storage (DACCS) is scientifically feasible and has delivered measured removals, but its contribution to mitigation at millions or billions of tonnes per year remains **unproven**. A public regional pilot may be justified as an *information-producing demonstration*, not as a near-term source of substantial climate mitigation. Funding should be conditional on independently verified **net durable removal**, genuinely low-emission energy without displacing more valuable decarbonization, credible and monitored storage, transparent full-chain costs, community safeguards, and a learning question that cannot be answered more cheaply.

The most recent internationally synthesized operational estimate is only about **1,500 tonnes of gross DACCO₂ stored in 2025**, drawn from registries for two Icelandic plants through October—not 1,500 net tonnes. An earlier estimate overstated 2023 activity by using a plant's maximum design capacity. Actual performance therefore matters much more than promotional nameplates. Independently reviewed engineering studies suggest costs can eventually fall, but estimates of **US$400–2,500 per gross tonne** across published analyses and an IEA **US$500–1,900 per tonne** estimate for current projects are neither fully harmonized nor verified market-wide expenditure. A future **US$100–600 per tonne** is conditional modelling, not a demonstrated price. Life-cycle emissions also matter: a 2026 synthesis reports **net/gross ratios of 23%–90%** in the limited available registry assessments of Orca and Mammoth. [1] [2] [4]

A defensible committee choice may be **targeted, staged pilot support**, **deferral** until energy/storage and measurement conditions are credible, or **priority for cheaper emissions prevention and alternative removals**. No international figure establishes that an unspecified region meets the conditions.

Scope, definitions and evidence method

This is a **structured narrative review**, not a systematic screening of all studies. It draws on peer-reviewed technology and life-cycle syntheses, the 2026 *State of Carbon Dioxide Removal*, the International Energy Agency, project registry-based reporting, and a recent operator disclosure. Searches targeted technology maturity, field output, energy/material requirements, costs, storage and funding risks, with contrary assessments considered. Evidence was assessed for **observed versus modelled** performance, source independence, numerator/denominator, system boundary, and relevance to an unspecified region. The 2025 systematic technology review examines literature largely through August 2024; the 2026 stocktake and operator update extend the operational picture. No independent, public full-chain cost audit of large-scale DACCS was established by this review. [1] [2] [3]

Definitions. *Direct air capture* (DAC) chemically separates CO_2 from ambient air. *DACCS* adds durable underground geological or mineral storage. *Gross captured* means tonnes separated from air; *gross stored* subtracts losses before storage; **net durable removal** additionally subtracts relevant life-cycle greenhouse gases and any reversal of storage. A tonne used in fuel, food or other short-lived products is normally re-emitted and is **not** a durable removal tonne. Storage and utilization must never be treated as interchangeable. [1] [5]

Q1 — What DAC technologies exist, what has been demonstrated operationally, and what remains unproven?

Answer (high confidence on chemistry; moderate on commercial maturity): Air separation and geological storage each work, and limited integrated DACCS operations are documented. Routine high utilization, long sorbent life, predictable whole-plant cost, and repeated large-scale operation are not yet established. Four principal families differ:

DAC family	How it releases captured CO_2	Evidence status and scaling question
Solid sorbents	Reusable solids bind CO_2; moderate heat, often roughly **80–120 °C**, releases it.	Commercially operating small plants; durability, humidity effects, fan energy and uptime remain consequential.
Liquid solvents	Alkaline liquid binds CO_2; some regeneration routes require heat near **900 °C**.	Technology family assessed at pilot/demonstration stage; heat supply, water and capital intensity critical.
Mineral/calcium looping	Reversible reactions with calcium- or magnesium-based solids.	Certain variants commercially demonstrated in limited configurations; materials handling and integration at high throughput uncertain.
Electrochemical and other emerging routes	Electricity or novel cycles regenerate capture materials.	Laboratory to pilot evidence; promised low energy/cost not yet a validated full-chain result.

*Table 1. Technology mechanisms and maturity; categories are broad and do not certify an individual design. The 2026 stocktake assigns technology-readiness levels around 9 to mature solid/mineral routes, 6–7 to liquid-solvent routes and 4–6 to electrochemical regeneration; these are **not** scores for gigatonne-scale deployment. [2] [3] [5]*

What actual operation shows. The 2026 *State of CDR* reconstructs **2025 DACCS gross flow to durable storage of 0.0015 $MtCO_2$ (1,500 t)** from published registry data for Climeworks' Orca and Mammoth, available to October 2025. Its earlier 2023 estimate of 0.004 Mt relied on Orca's **4,000 t/yr nameplate**, not its achieved output; revised data suggest the actual 2023 flow was about one eighteenth of the 2025 value. The sites were still ramping below nameplate in 2025; another announced facility without traceable verification was excluded. This is more convincing than a company announcement, although based on registry reporting rather than a cross-industry independent plant census. [1]

A newer, weaker-evidence signal: in September 2026 Climeworks claimed Mammoth delivered **675 net tonnes in January–June 2026**, versus **119** in the same 2025 period, after collector upgrades. The two upgraded containers reportedly improved markedly, but the company disclosed **neither audited plant-wide cost per net tonne nor evidence that all equipment sustained the upgraded performance**. This vendor disclosure is evidence of a plausible learning direction, not independent proof that Mammoth's **36,000 t/yr design capacity** is reached. [7]

Q2 — What limits expansion to climate-relevant scales, including energy, materials, infrastructure and permanent CO_2 storage?

Answer (high confidence that constraints matter; low confidence in a global deployable maximum): The limiting resource is not ambient CO_2 but coordinated

low-carbon electricity and heat, reliable capture media, buildable equipment, CO_2 transport, injection capacity and permissions. Modelled global potential is not a schedule of buildable projects.

Energy at scale. The WRI uses an *illustrative* future DAC energy assumption of **2,000 kWh per tonne**, not a measured value for every technology; some systems need both electricity and heat and should not be represented as using 2,000 kWh of grid electricity. At that same illustrative energy-equivalent intensity, **1 million t/year × 2,000 kWh/t = 2 TWh/year**; **1 billion t/year = 2,000 TWh/year**. At ordinary industrial scale that competes with electrification and fossil power displacement unless additional clean generation is genuinely available. The peer-reviewed synthesis finds no clear historical trend of declining published DAC energy requirements; large improvement is an engineering goal, not a demonstrated fleet result. [3] [5]

A source may be contractually "renewable" yet not deliver the climate benefit assumed if its capacity would otherwise displace emitting generation. For evaluation, distinguish **plant accounting emissions** (the electricity purchased) from **system consequences** (which generation or abatement is displaced, including across hours). An energy source with low average emissions is necessary for strong net removal but does not establish zero opportunity cost. [3] [5]

Materials, sites and storage. Sorbents/solvents degrade, require replacement, and have upstream emissions; air humidity and temperature affect capture. Fans, contactors, pumps, kilns, heat exchangers and CO_2 conditioning need manufacturing and reliable operations. Water demand, chemical handling, land for associated power supply and local pollution require site-specific assessment—some solvent designs consume water whereas some sorbent systems may recover it. [3] [5] [6]

Geological storage has decades of experience at project scale and, if well chosen and monitored, low expected leakage; basalt mineralization can lock carbon into rock. But the theoretical geological capacity of **10,000+ $GtCO_2$** is not accessible storage on demand: the 2026 assessment suggests perhaps **~1,000 Gt** usable after physical and social constraints. Near any proposed region, **injectivity** (the annual rate that can safely be injected), connected transport, groundwater protection, fault/pressure assessment, monitoring, liability and public consent could be binding before total volume is. [2]

Reality check on forecasts. Facilities operating or under construction imply only **~0.55 Mt/yr of potential DACCS capacity by 2030** *if all reach rated output*, according to the 2026 stocktake; financing, permitting and verification can reduce it. Its high-ambition scenario family spans **<0.1–1.8 Gt/yr DACCS by 2050**, a range of model outcomes—not measured progress or an agreed forecast. The larger of those endpoints would require unprecedented expansion in clean energy, manufacturing, CO_2 storage services and workforce. [1] [2]

Q3 — What are credible current and projected removal costs, and how sensitive are they to technological and economic assumptions?

Answer (moderate confidence in the present order of magnitude; low in future point estimates): Costs remain high and apples-to-apples comparisons are unusually difficult. Every quote should specify year, currency, gross or net denominator, annual utilization and whether compression, storage, monitoring, financing and developer costs are included.

Evidence type	Estimate (US$/tCO$_2$)	What the number does—and does not—say
IEA 2025, **estimated current projects**	**500–1,900**	Institutional estimate, not publicly audited delivered cost for all plants; denominator and boundaries need project validation.
2025 peer-reviewed engineering studies, compiled in 2026	**400–2,500 / gross captured**	Heterogeneous *modelled levelized capture* cost; many omit parts of the full value chain.
Optimistic best-design model, published 2025	**97–168 / gross**; **126–170 / net**	Conditional simulation under favourable technical assumptions, **not an observed project price**.
Learning/scale projection in 2026 synthesis	**100–600 / gross**	Conditional gigatonne-scale projection, highly sensitive to learning and energy assumptions.

*Table 2. Estimates are deliberately **not averaged** because the denominator, system boundary, maturity and model assumptions differ. IEA also proposes roughly **US$300/t by mid-century** as a possible future level, not a guarantee. The 2025 systematic review sometimes expresses its forecast using a net denominator; that value is not interchangeable with the gross range summarized in the 2026 chapter. [2] [3] [4] [11]*

Why the spread is real. Capture performance per unit of sorbent and its replacement lifetime drive equipment quantity; electricity/heat cost and carbon intensity drive annual operating and carbon-accounting costs; financing cost, depreciation, capacity factor and unplanned outages change the denominator. The 2025 systematic review found a scarcity of comparable geographically diverse cost studies, and often-missing owner costs, contingencies or transport/storage charges. These explain why headline figures can move without genuine technological learning. [3]

Geological **compression alone** is estimated around **US$10–30/t**, while onshore storage may range from a few dollars to **<US$25/t** and offshore storage roughly **US$5–55/t**, contingent on site and scale. Transport can add materially more. These figures come from different system boundaries and should **not** automatically be added to estimates already including such items. [2]

Simple denominator test (hypothetical, not measured). Suppose a fully specified DACCS system costs **$800 per gross tonne captured**, and storage/life-cycle deductions leave **80%** net durable removal. Then **$800 ÷ 0.80 = $1,000 per net tonne**. If only **50%** remains net, the identical gross capture cost becomes **$1,600 per net tonne**. An investment comparison using $800 while an alternative is priced per *net* tonne is misleading; further cost changes from storage or delays would increase the difference. [1] [3]

Q4 — How much CO_2 is durably removed after accounting for life-cycle emissions, energy requirements and storage risks?

Answer (high confidence in accounting principle; low in transferable field ratio): Verified climate benefit is **not** captured volume. A project has positive net removal only when storage is real and sustained and its cradle-to-grave greenhouse-gas footprint is smaller than the atmospheric CO_2 stored.

> **Net durable CO_2 removed = metered atmospheric CO_2 durably stored − life-cycle GHG emissions (CO_2e) − expected reversal/leakage adjustment.** The accounting period, permanence horizon, treatment of upstream electricity and materials, and uncertainty deductions must be declared. Avoid double-counting stored CO_2 or avoided fossil emissions.

The 2026 *State of CDR* reports **99%–100% of gross captured CO_2 transferred into durable storage** for the limited assessed DACCS records, but just **23%–90% net removal per gross captured tonne**, due largely to lifecycle emissions and varying registry boundaries at **Orca and Mammoth**, both powered by relatively low-carbon geothermal energy. This is **not** a universal performance interval, and the study's 2025 **1,500 gross tonnes must not be relabelled 1,500 net tonnes**. The wide spread despite favourable energy conditions makes independent full-chain metering a prerequisite for funding. [1]

Peer-reviewed life-cycle synthesis includes scenarios in which grid-powered DAC has near-zero or even **positive** net emissions, whereas renewable-powered cases are more strongly negative, depending on process and system boundary. Conversely, a **2026 prospective** calcium-looping LCA using industrial inputs models **85%–96% net efficiency in 2050**, especially with clean power and integration; these are plausible technological countercases, **not measurements from a 2050 commercial fleet**. Neither pessimistic nor optimistic cases substitute for local verified energy and material inventories. [6] [10]

Evidence should include calibrated intake/output CO_2 measurements; separately metered electricity, heat and water; baseline and actual material replacement; construction, transport, compression, storage and monitoring emissions; records of injected quantity and storage integrity; uncertainty bounds and third-party audit. Storage credits must be backed by continuing monitoring, funded remediation and long-term responsibility—not merely a CO_2 sales contract. Using DAC-derived CO_2 for synthetic fuels may substitute for fossil fuels but is **not durable removal** when burned; enhanced oil recovery also introduces fossil-production and accounting risks. [1] [2] [5]

Q5 — What can pilot projects realistically establish, and what evidence, safeguards and performance milestones should determine whether public funding is warranted?

Answer (high confidence in testable-versus-system distinction; committee thresholds are proposed governance conditions): A carefully designed pilot can reduce uncertainty about a particular configuration's **seasonal reliability, energy use, sorbent degradation, upstream carbon balance, measured storage, full-chain cost and local acceptability**. It cannot, by itself, validate a million-tonne factory, a gigatonne supply chain, widely available additional clean power, national-scale pipeline networks or future unit costs. Capturing CO_2 in a test unit without storing it does not certify durable removal. [3] [4]

Proposed performance contract: demand measured **net durable tonnes**, **MWh of electricity and GJ of heat per gross and net tonne**, CO_2 storage mass balance, annual availability/uptime, sorbent/chemical replacement rate, **fully loaded US$ (or local currency) per net tonne**, data completeness and measurement uncertainty. Pre-agree a baseline and independently reviewed pass/fail targets for each indicator—including a **strictly positive net-removal result** under relevant energy accounting—rather than imposing a universal numerical cost or energy threshold without regional evidence. A **full operating year** is a prudent proposed minimum before claims of seasonal reliability; it remains insufficient for decades-long storage certainty.

Stage / committee action	Evidence that should unlock the next tranche	Failure or stop condition
Pre-award: fund design/ testbed only	Published testable learning question; transparent comparator to emissions prevention/ other removals; credible additional clean energy; storage capacity and responsible operator plausibly secured; community consultation and safety plan.	Claimed reductions rely on short-lived CO_2 utilization, unspecified storage, or electricity that would foreseeably displace higher-benefit abatement.
Commissioning: conditional pilot construction	Independently checked capture/storage measurement plan, cradle-to-grave LCA protocol, financial and operating data rights, water/chemical and CO_2 transport safeguards, binding monitoring and remediation finance.	Unable to measure net output or obtain a feasible monitored storage route; material community or environmental risks unresolved.
Operation: pay for verified learning and output	At least one full seasonal cycle of auditable CO_2 mass balance, real power and heat use, uptime, consumables, complete cost ledger, uncertainty reporting and positive net tonnes; public non-confidential results, failures included.	Net climate result ≤0, missing data, unreported safety incidents, or performance outside pre-agreed tolerances without a credible corrective case.
Replication/scale: separate approval	Demonstrated improvement on prespecified targets, cost per *net* tonne under conservative operating assumptions, secured larger storage injection rate, power-network counterfactual and demand/offtake beyond public subsidy.	Promising unit performance is generalized to region-wide buildout without energy/storage and financing proof.

Table 3. Proposed stage gates are decision rules, not observed technical norms or region-independent law. Each stage should have capped public exposure, transparent third-party assurance, an exit right and shared learning so benefits survive failure. Private co-funding can align incentives but does not replace proof. [4] [8]

Opportunity cost and institutional safeguards. Judge a pilot on *additional information and options created per public euro or dollar*, separately from immediate removals delivered. Include a counterfactual portfolio: feasible near-term emission reductions and other CDR with similar **netness, durability, monitoring and risk**, rather than making a simplistic dollar-per-ton ranking of unlike activities. The IEA notes first-of-a-kind project support and open-access testbeds can generate public learning; one large purchaser accounted for approximately **65% of 2024 CDR credit purchases**, creating off-take concentration risk. A more sceptical Project Drawdown assessment rejects DAC deployment as a priority because of current low output, resource and financing opportunity costs, and potential fossil-fuel lock-in; these concerns should be tested, not dismissed. [4] [9]

Q0 — Under what technical, economic, environmental and institutional conditions could direct air capture achieve a meaningful contribution to climate change mitigation, and what evidence would justify public investment in regional pilot projects?

Conditional judgment. DACCS could make a meaningful **complementary** contribution if successive plant generations reliably achieve much higher utilization and lower *full-chain cost per net tonne*, the needed clean energy is built without delaying emissions cuts, materials and water impacts remain acceptable, and verified multi-decade geological storage and institutions expand together. The scale of the need is not a license to plan as if the capacity already exists. A published model of climate planning finds that assuming high future CDR and then failing to deploy it can aggravate fossil lock-in and later adjustment costs; this is a **scenario-based warning**, not a forecast. [2] [8]

Support a limited pilot when it answers a material and transferable uncertainty, has credible local low-carbon energy plus storage, independent measurement and publication of failures, pre-set stage gates, and a better risk-adjusted learning case than alternatives. **Defer** when site power additionality, storage or monitoring cannot yet be evidenced, but those gaps are remediable. **Deprioritize** when the proposed funding is mainly justified by speculative nameplate removals, unverifiable credits, high indirect emissions, unresolved hazards, or an opportunity cost strongly favouring available emission reductions. Even a successful pilot **does not authorize commercial-scale rollout** without a new region-specific appraisal.

Limits and research conclusion

Evidence is unequally strong: first-principles chemistry and actual **small gross stored volumes** are well supported; **delivered net totals, full-chain audited plant costs, site-to-site reproducibility and region-specific consequences** are thinner. The 2026 stocktake's net-efficiency range uses two registry-documented Icelandic facilities and non-identical life-cycle boundaries. IEA cost ranges and prospective peer-reviewed engineering models are not directly comparable; company operating updates remain self-reported, even when consistent with a plausible improving trajectory. New technologies' optimistic LCAs do not establish factory performance, while early critics extrapolating extremely large DAC portfolios also cannot rule out a useful smaller role. [1] [2] [3] [7] [12]

No local electricity dispatch, permitting regime, storage reservoir, water availability, budget, vendor or candidate project was specified. Accordingly **no regional facility is endorsed or rejected**. Research was closed when Q0–Q5 could be answered with appropriately qualified international evidence and genuine countercases; additional global searches would be unlikely to resolve the locally dependent funding decision. The central public-value proposition is **verified, shareable learning under strict climate and fiscal safeguards**, not the assumption that building a pilot delivers climate-relevant removal.

References

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